

Lenalidomide

Lenalidomide is a cancer drug. It is used to treat myeloma, some types of lymphoma and myelodysplastic syndromes (MDS).

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What is lenalidomide?

Lenalidomide is used to treat:

- [myeloma](#)
- [follicular lymphoma](#)
- [mantle cell lymphoma](#)
- [myelodysplastic syndromes \(MDS\)](#).

Lenalidomide is a type of targeted therapy drug and also an immunotherapy drug. It works by:

- stopping the growth of cancer cells (cancer growth inhibitor)
- helping the immune system attack and destroy cancer cells ((immune system modulator)
- blocking the signals that cancer cells need to make new blood vessels, which can slow the growth of the cancer (angiogenesis inhibitor).

It is best to read this information with our general information about [targeted therapy drugs](#) and the [type of cancer](#) you have.

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

Having lenalidomide

Lenalidomide comes in capsules. This means you can take it at home. You may have it with a [steroid](#) drug called dexamethasone or with other cancer drugs.

During a course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your targeted therapy.

Your course of treatment

You usually have a course of several cycles of treatment over a few months. Your cancer team will discuss your treatment plan with you. How often you take lenalidomide will depend on the type of cancer you have and whether you are having it with other cancer drugs.

Your cancer team may also give you anti-sickness drugs and other medicines to take home. Take these exactly as they tell you to.

Taking lenalidomide capsules

Your cancer team will give you the capsules to take home. They may give you capsules of different strengths.

You can take lenalidomide with or without food. Always take the capsules exactly as explained. This is important to make sure they work as well as possible for you.

Make sure you:

- swallow them whole with a glass of water
- do not chew, open, or crush them
- take them at the same time every day.

If you forget to take lenalidomide and less than 12 hours have passed, take the dose as soon as possible. If more than 12 hours have passed, do not take the missed dose. Just take your usual dose at the usual time the next day. Do not take a double dose to make up for the one you missed. Contact

the hospital on the 24-hour number you have been given for advice.

Other things to remember about your capsules:

- Wash your hands after taking them.
- Other people should avoid direct contact with them. If possible, you should handle the capsules yourself. If you need help, the person helping you should wear disposable gloves. If pregnant, the person should not touch or handle them at all.
- Keep them in the original package and at room temperature, away from moisture, heat and direct sunlight.
- Keep them safe, where children cannot see or reach them.
- If you are sick (vomit) just after taking them, do not take an extra dose. Take your next dose at the usual time.
- Do not throw away unused capsules. Return them to your cancer team at the hospital.

Preventing pregnancy while taking lenalidomide

You must not [become pregnant or make someone pregnant](#) while taking lenalidomide. This is because it may cause severe abnormalities in developing babies.

You will take part in a Pregnancy Prevention Programme during treatment. Even if you do not think you are at risk of getting pregnant or making someone pregnant, you will be included in this programme.

Your doctor, nurse or pharmacist will give you information about the programme and the risks of lenalidomide and pregnancy. They will ask you to sign a [consent form](#) to show you understand the information.

Pregnancy tests and contraception

If you are of childbearing age and able to get pregnant, you will have a pregnancy test before starting this treatment. The pregnancy test is repeated every 4 weeks during treatment, and 4 weeks after treatment finishes.

You must use an effective form of contraception, such as an implant, injection or the progesterone-only pill. You should not use the combined oral contraceptive pill. It increases your risk of developing blood clots.

You must use contraception to prevent pregnancy:

- for 4 weeks before treatment
- during treatment
- for 4 weeks after treatment finishes.

If you think you may be pregnant at any time during your treatment, contact your doctor, nurse or pharmacist straight away.

Contraception to prevent making someone pregnant

Lenalidomide can pass into the semen that carries sperm. During treatment and for a week after treatment finishes, you must not:

- make someone pregnant
- donate sperm.

You can protect partners by using a condom during sex. During treatment and for a week after treatment finishes, you must use a condom during sex with anyone who is:

- pregnant
- of childbearing age and able to get pregnant.

If a partner thinks they might be pregnant during your treatment, contact your doctor, nurse or pharmacist straight away.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Most common side effects

These side effects are likely to be the most common side effects for this treatment. They happen to 10 or more people in every 100 people (10% or more) who have the individual drug. It can also depend on your situation and the type of cancer you are being treated for.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

You may be given a drug called [G-CSF](#). This encourages the body to make more white blood cells. You have it as a small injection under the skin.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums

- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Feeling sick

You may feel sick during this treatment, but this is usually mild. Your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. [If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Skin changes

This treatment may cause inflammation in the skin. This is usually mild, but if it is not treated it can become serious. You may need creams, steroids or other medicines to treat skin problems.

Contact the hospital on the 24-hour number as soon as possible, if you notice any skin changes during treatment, or after treatment ends. This includes:

- a rash or bumps on the skin
- dry, tight or itchy skin
- areas that are a different colour to your usual skin tone.

Here are some things you can do to take care of your skin during treatment:

- Wash with lukewarm water and mild, unperfumed, soap-free cleansers.
- Moisturise your skin regularly with unperfumed moisturisers for dry skin conditions.
- Use sun cream of at least SPF 30 to protect your skin from the sun. Cover up with clothing and a hat.

Contact the hospital straight away on the 24-hour number if you have any of these serious skin symptoms:

- a rash that is spreading
- swollen, blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Headaches or dizziness

Let your doctor, nurse or pharmacist know if you have headaches or feel dizzy. If you feel dizzy, do not drive or use machinery.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Mood changes

You may have mood changes. Some people may become low in mood or depressed. Talk to your doctor, nurse or pharmacist if you are having any of these difficulties. They can give you advice and support.

Muscle or joint pain

You may get pain in your muscles, joints or in your back. Some people may have pain in the area of the cancer. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable. Having warm baths and resting regularly may help.

Tell your doctor, nurse or pharmacist if the pain does not get better.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Changes in thyroid function

This treatment may affect how well the thyroid gland works. It may sometimes cause an under active thyroid where the thyroid gland does not produce enough hormones. This is called hypothyroidism. It can cause symptoms such as tiredness and weight gain.

Effects on the liver

This treatment may affect how your liver works. It can cause liver inflammation called hepatitis. You will have regular blood tests to check how your liver is working. Tell your doctor, nurse or pharmacist if you have loss of appetite.

Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen).

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Effects on the kidneys

This treatment can affect how the kidneys work. This is usually mild and goes back to normal after treatment ends. You will have blood tests to check how well your kidneys are working. Contact the hospital on the 24-hour number if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Drinking fluids helps protect your kidneys. The advice is usually to try to drink at least 2 litres (3½ pints) of fluid each day. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

We have more information about [coping with changes to taste](#).

Dry mouth

You may get a dry mouth. Drink plenty of fluids and add sauces and gravies to your food to keep it moist. If you have dry lips, using a lip balm can help.

Eye problems

Tell your doctor, nurse or pharmacist if:

- you have blurry vision during treatment
- your eyesight changes.

Blood pressure changes

This treatment may cause low or high blood pressure. Tell your doctor, nurse or pharmacist if you have ever had any problems with your blood pressure. Let them know if you feel dizzy or have any headaches. You will have regular blood pressure checks during treatment.

Falls

You may be more likely to have a fall when taking this treatment. Try to make sure areas you walk through are well lit, free of clutter and remove things you could fall over.

Allergic reaction

This treatment might cause an allergic reaction. But this is not common. Signs of a reaction can include:

- a skin rash or itching
- feeling breathless or wheezy
- swelling of your face, mouth or throat.

If you feel unwell or have any of these signs, contact the hospital straight away on the 24-hour number. Do not take any more of this treatment until you have checked with them.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker

- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Effects on the nervous system

Rarely, this treatment can affect the nervous system. It can cause a serious condition called progressive multifocal leukoencephalopathy (PML). Symptoms of this include:

- blurred vision, loss of vision or double vision
- difficulty speaking
- weakness in an arm or a leg
- a change in the way you walk or problems with your balance
- persistent numbness
- decreased sensation or loss of sensation
- memory loss or confusion.

If you have any of these symptoms or someone with you notices you have these symptoms it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away.

Second cancer

This treatment can increase the risk of developing a second cancer years later. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can give you more information.

Other important information

Lactose

This treatment may contain lactose. If you have a lactose allergy or intolerance, ask your doctor, nurse or pharmacist for more information.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Hepatitis B and shingles

Some viral infections can become active again while you are taking this treatment. This includes hepatitis B (a liver infection) and herpes zoster (shingles). These may need treatment with anti-viral drugs. Your doctor or nurse will talk to you about this. Before you start treatment, you will have a blood test to check if you have had hepatitis B.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help